UVEN 2 mk3 – Assembly Instructions

Version: v1.0
Date: 29.11.2025
Author: Simon Trendel

1. Purpose & Scope

This document provides comprehensive instructions for assembling the UVEN 2 mk3 device. It is intended for technicians, engineers, and users responsible for the setup and initial operation of the equipment. The scope includes:

- An overview of required tools and materials
- Step-by-step mechanical and electrical assembly procedures
- Safety guidelines and recommended personal protective equipment (PPE)
- Final assembly checks and functional testing
- Revision history for tracking updates

By following these instructions, users will ensure correct and safe assembly, optimal device performance, and compliance with safety standards.

2. Safety Information

- Warning: [Example warning]
- Caution: [Example caution]
- PPE: [Safety glasses, gloves]

3. Required Tools

Tool	Notes

4. Bill of Materials (BOM)

uven2-mk3.xlsx

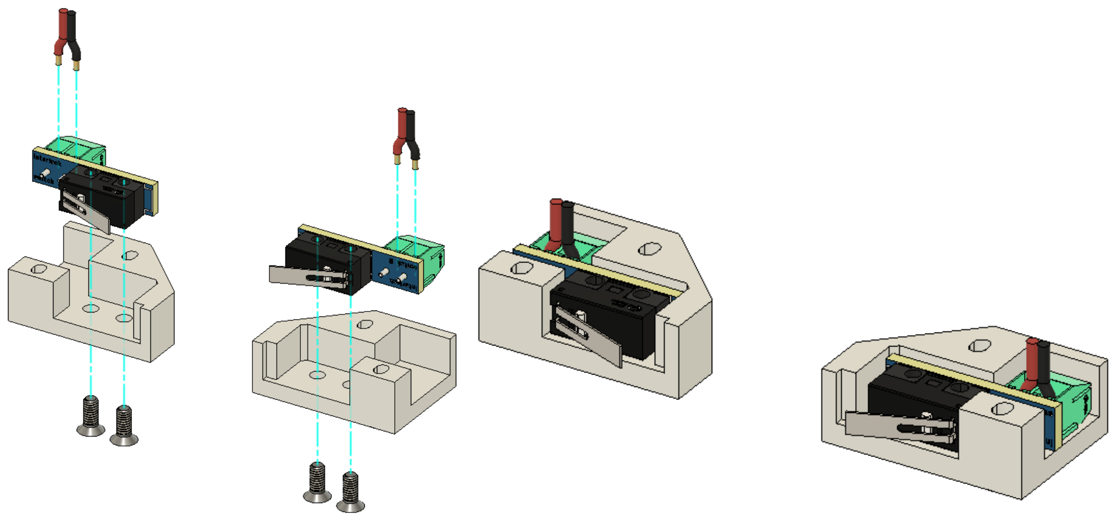
5. Assembly Overview

1. [Prepare workspace]
2. [Assemble base]
3. [Install wiring]
4. [Close enclosure]

6. Step-by-Step Assembly Instructions

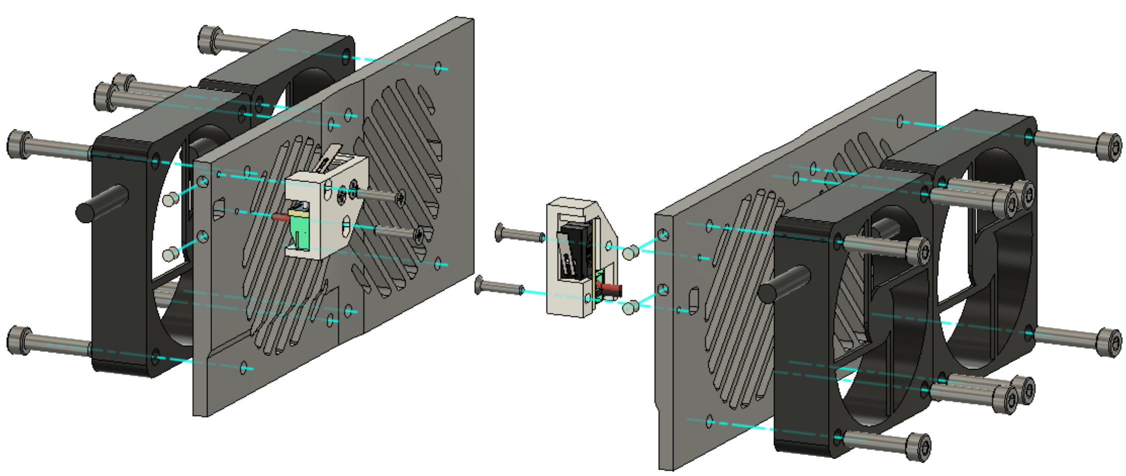
Step 1: Interlock switches

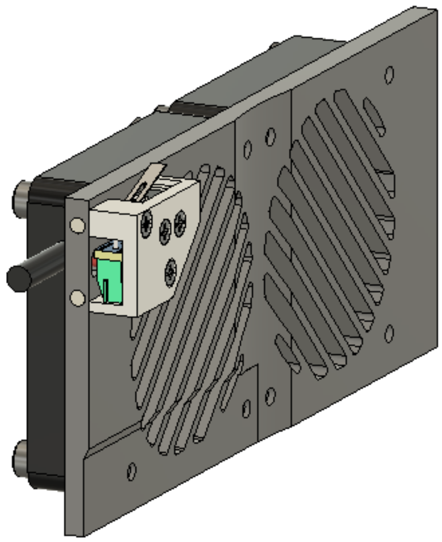
Goal: Connect the cables to the interlock switches. Mount the interlock switches into their covers. The cables need to be sufficiently long for later connection. Polarity does not matter.

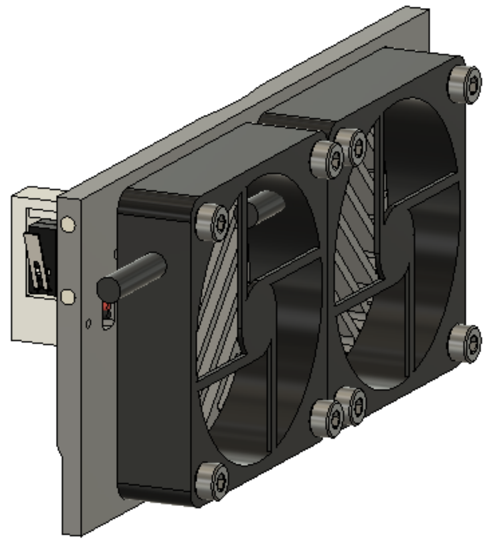


Step 2: Chamber fans, interlock switches, Magnets

Goal: Mount the 4 chamber fans using 16 x M4x18mm screws. Glue the magnets in. Mount the interlock switches from Step1 on either side using 2 x M2.5x12mm screws each. Glue in the 4 magnets.

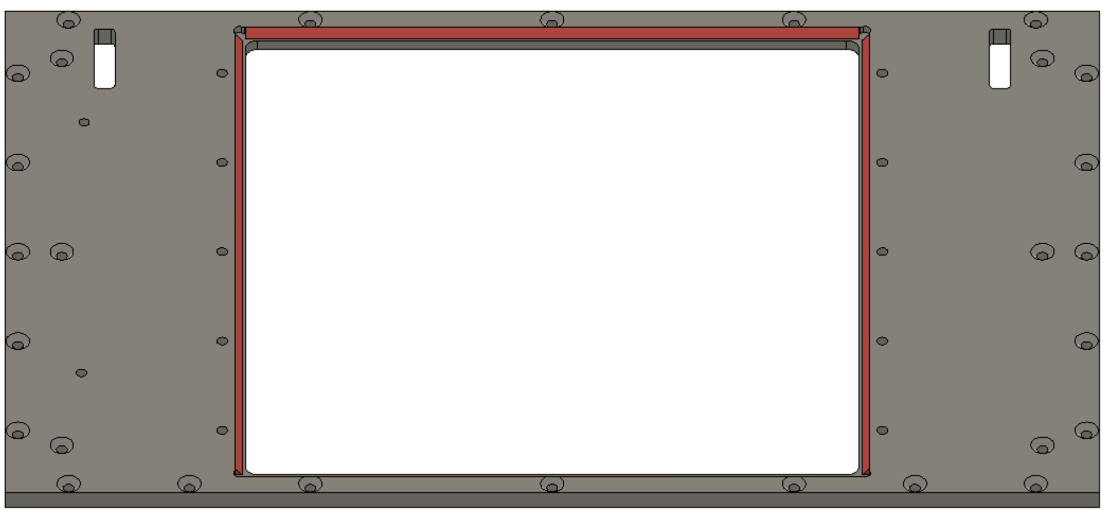


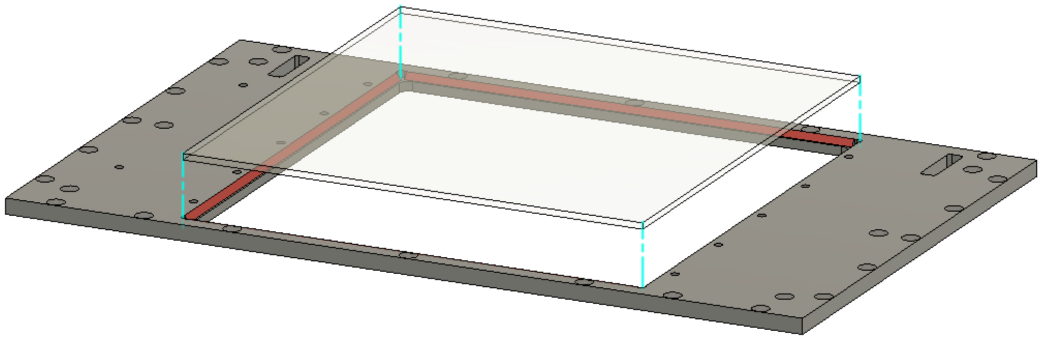


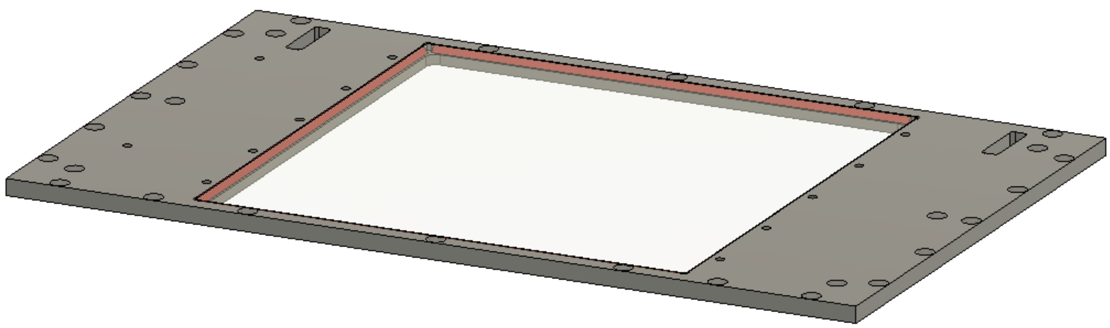


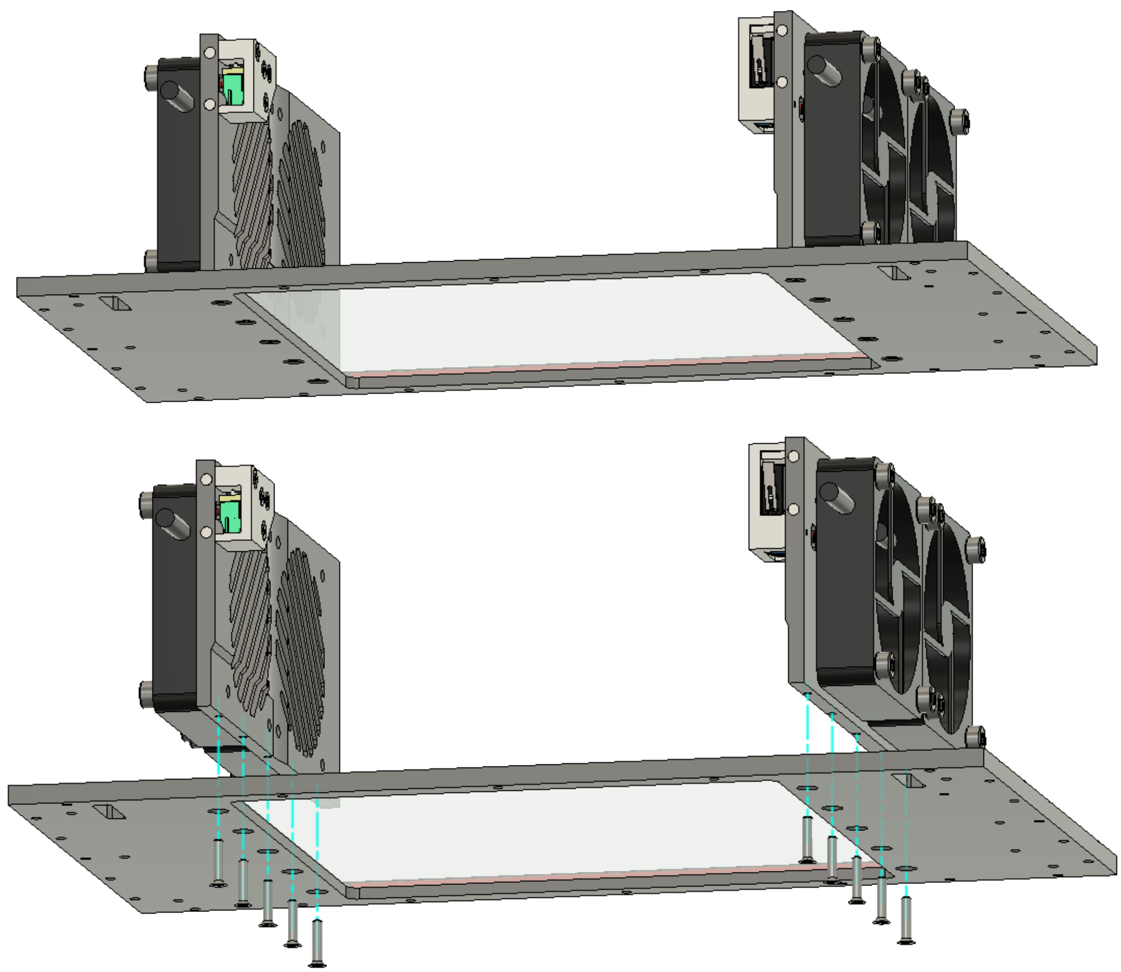
Step 3: Quartz glass, chamber sides

Goal: Glue in the quartz plate. Mount the chamber sides.



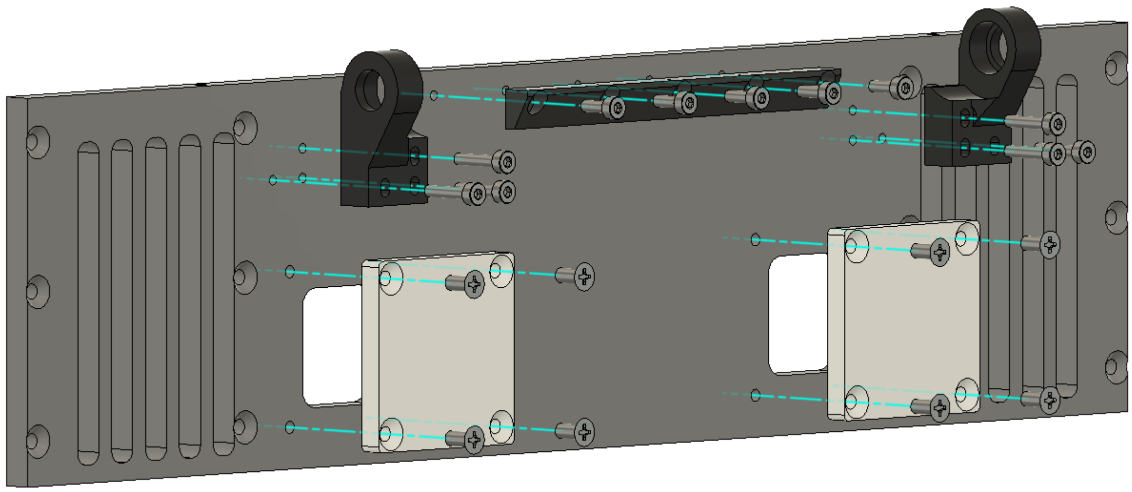


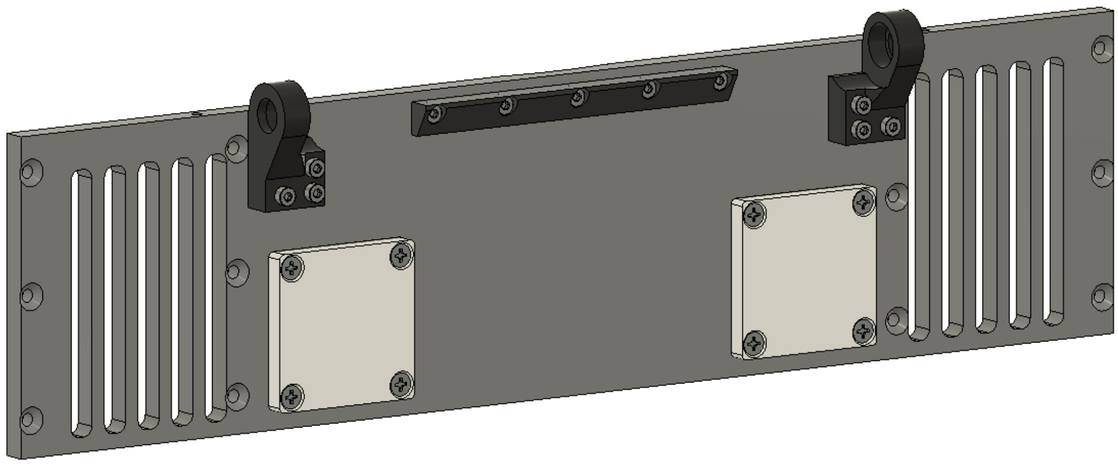




Step 4: Chamber plugs, hinges, endstop

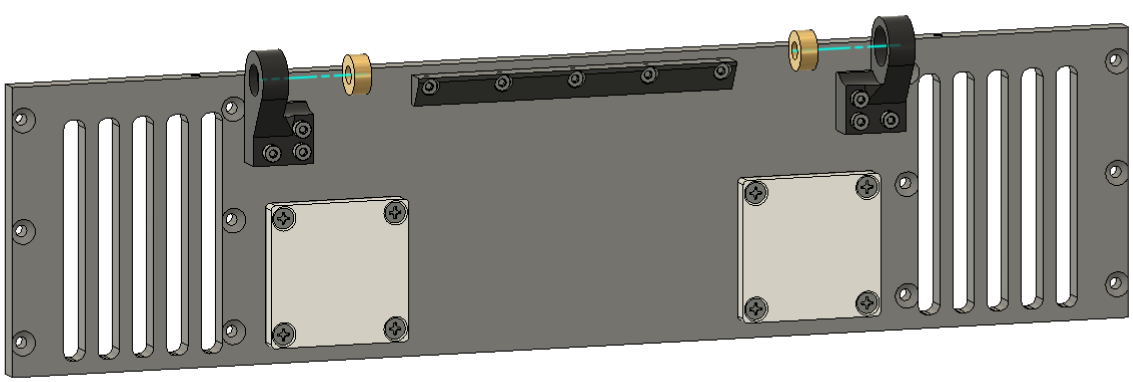
Goal: Mount the chamber plugs using 8 x M2.5x12mm screws. Mount the hinges using 6 x M2x10mm screws. Mount the endstop using 5 x M2x6mm screws.

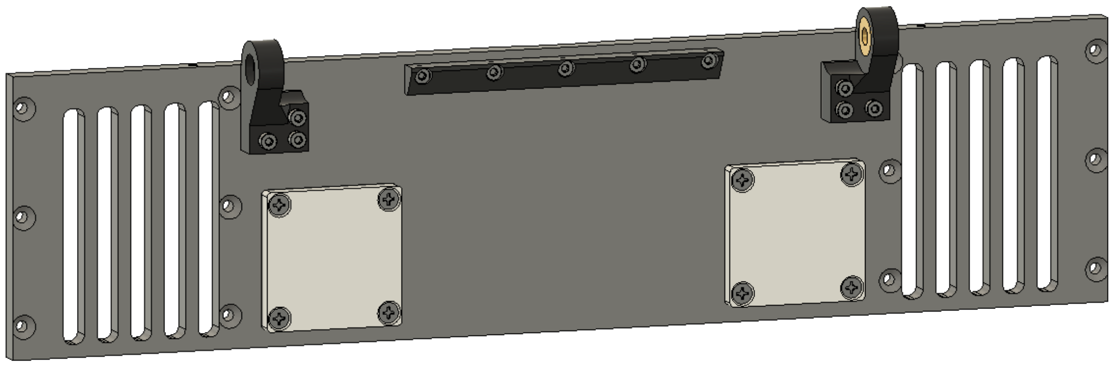




Step 5: Bearing/Bushing

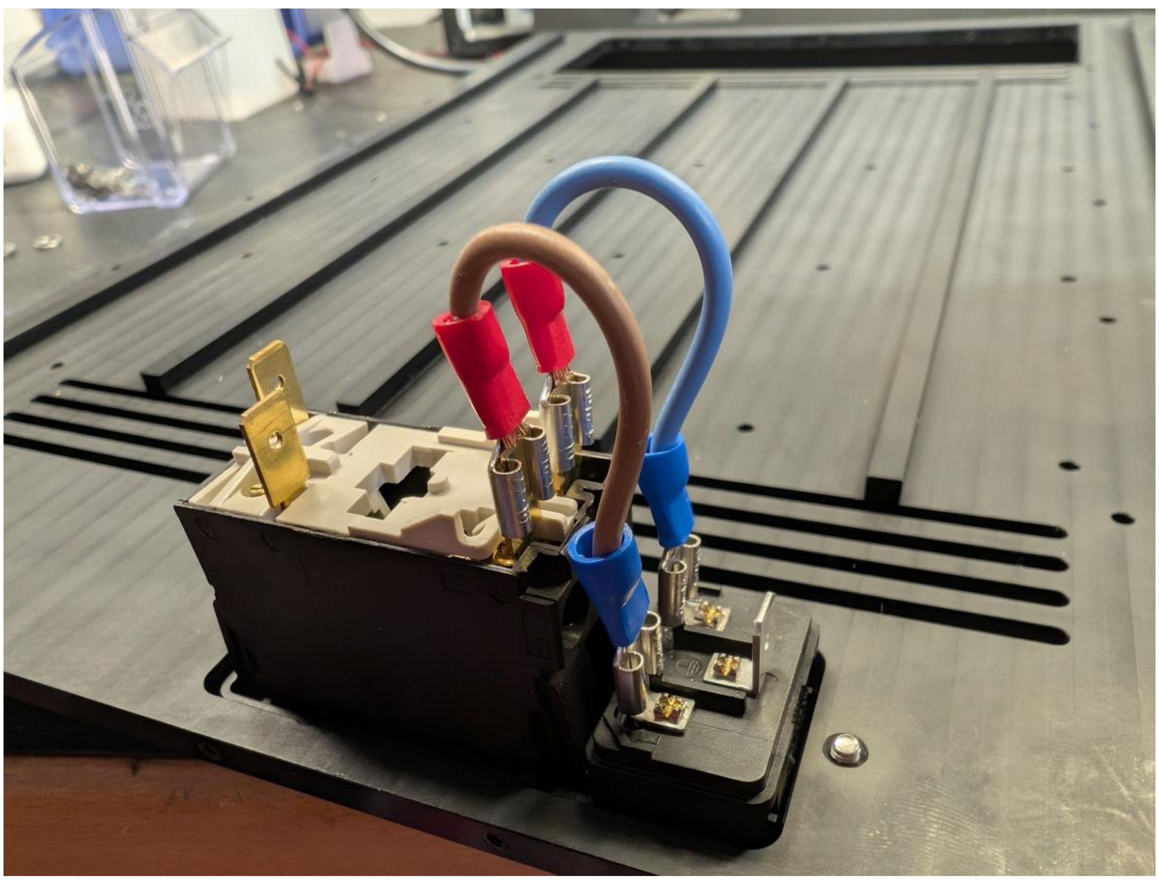
Goal: Press fit the bearing/bushing.

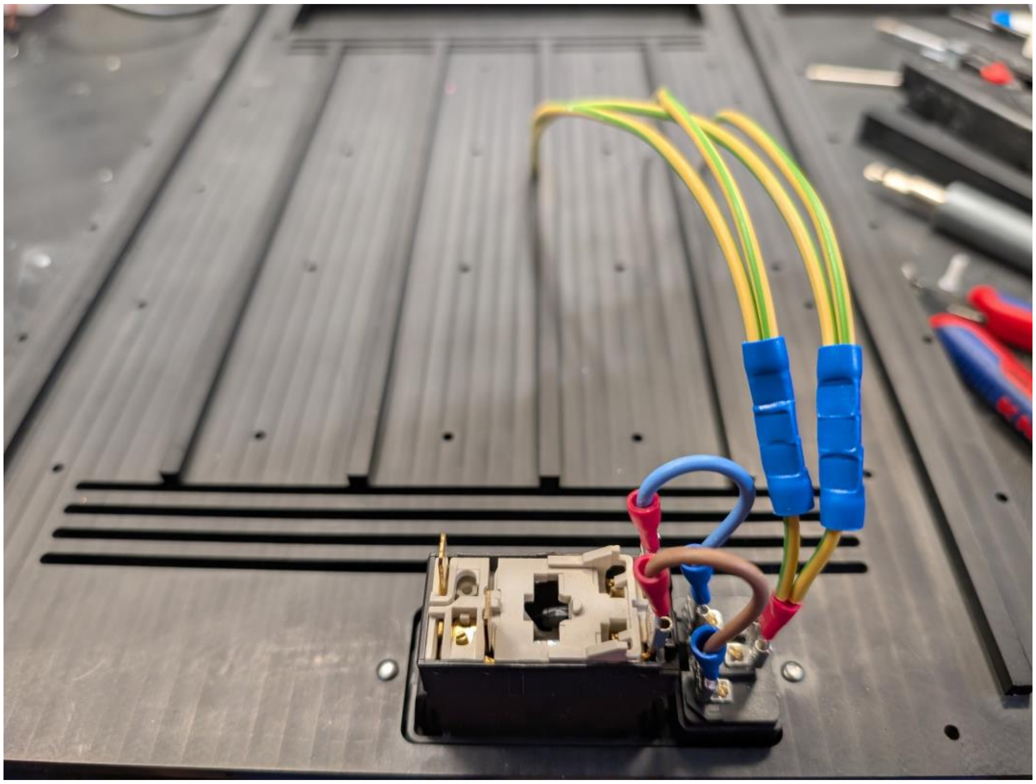


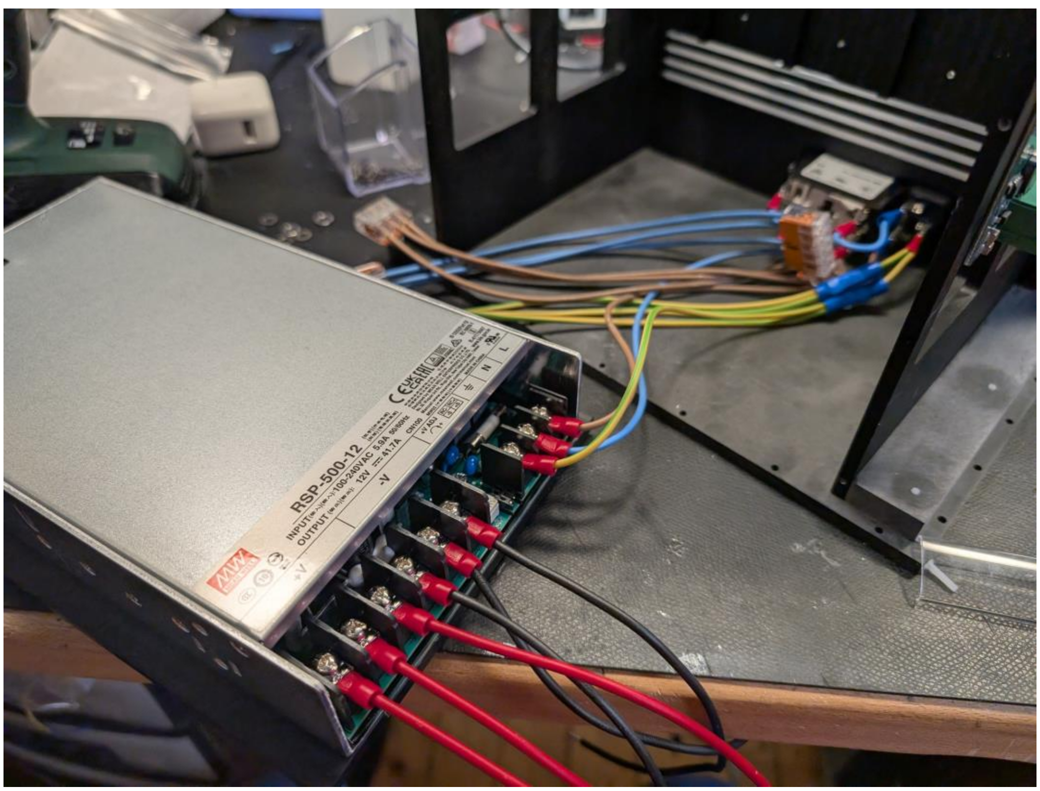


Step 6: Power switch, 12V power supplies

Goal: Mount the power switch using 2 x M3x8mm screws. Connect plug to switch (if not already connected). Attach GND, L, N cables. Use Wago 5x splitter to distribute four sets (one for each of the four 12V power supplies). Connect all four 12V power supplies to one of the power line sets. Connect the 12V power supply output cables 3xGND(black) and 3x12V(red) cables each. The cables should be long enough to reach one LED driver each.

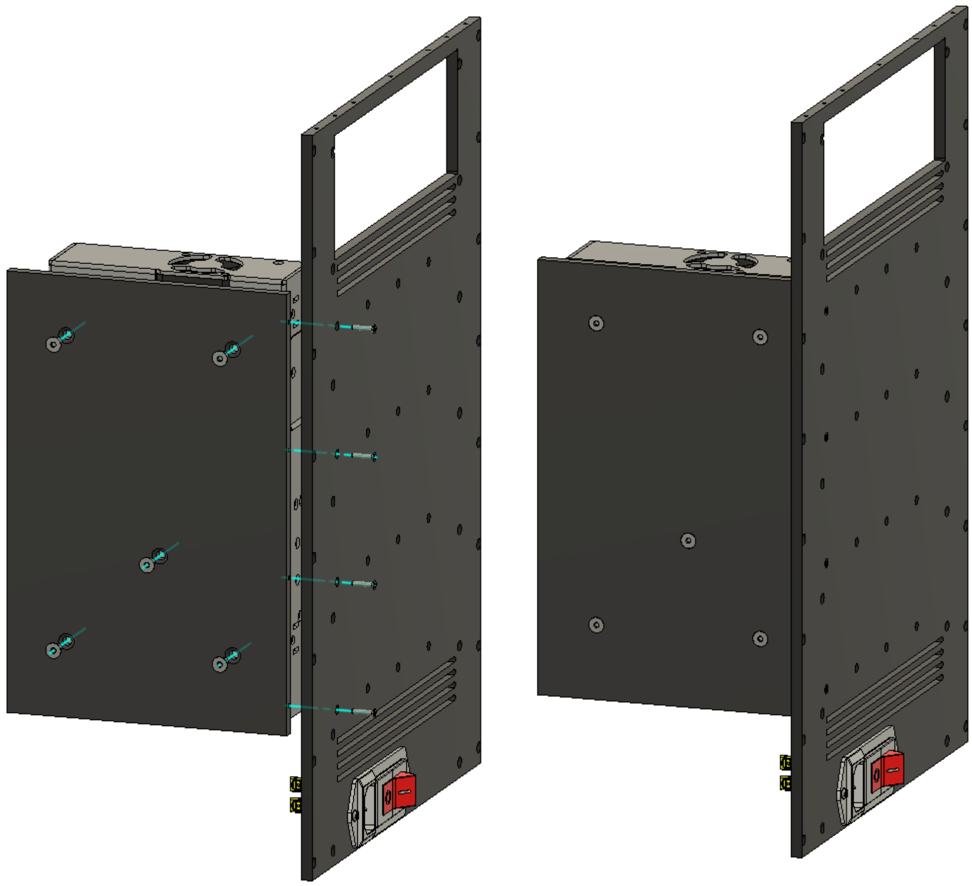


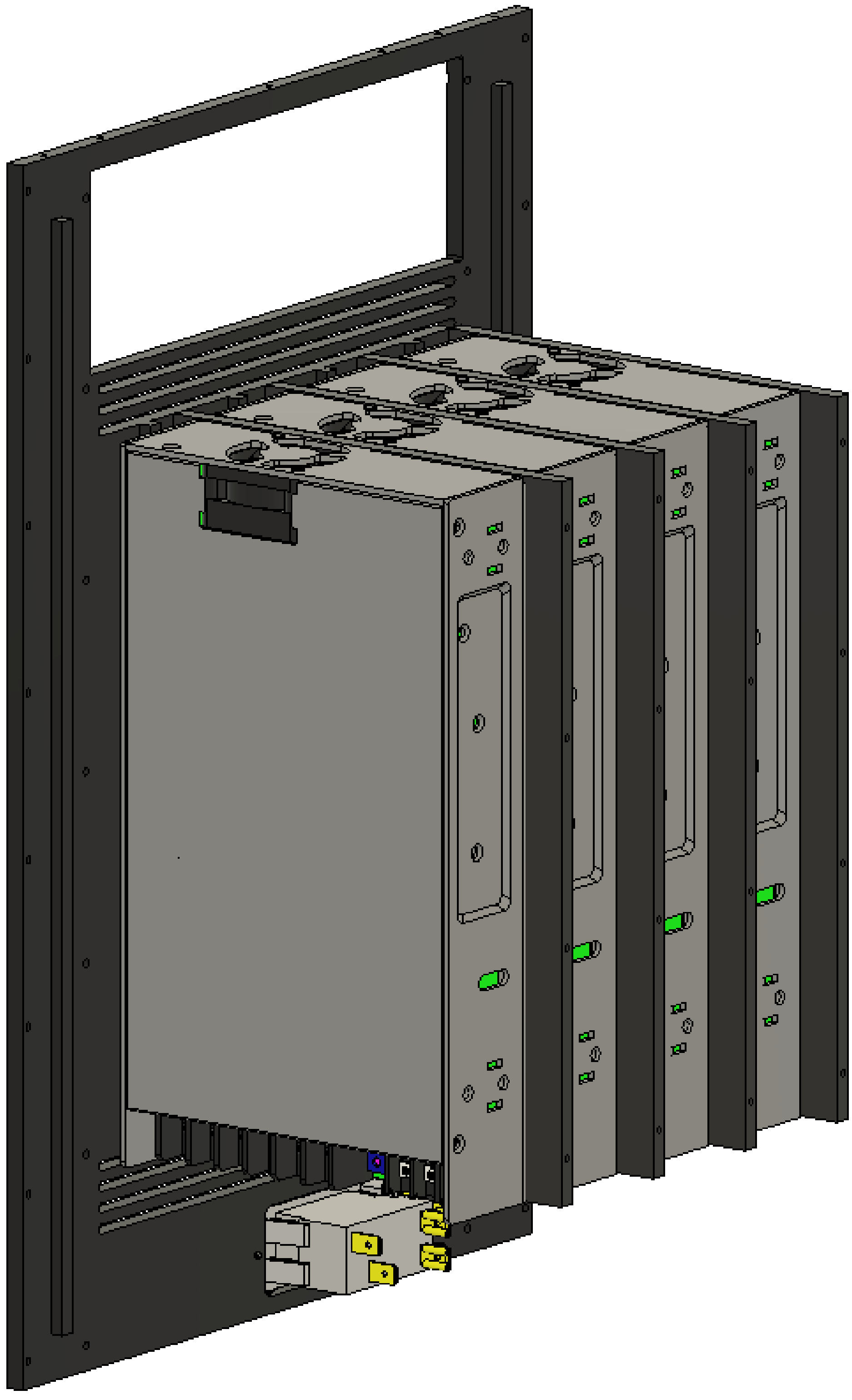



RSP-500-12
INPUT 100-240VAC 5.9A 50/60Hz
OUTPUT 12V 41.7A
+V
-V
+V ADJ
N
L

Step 7: Power supplies

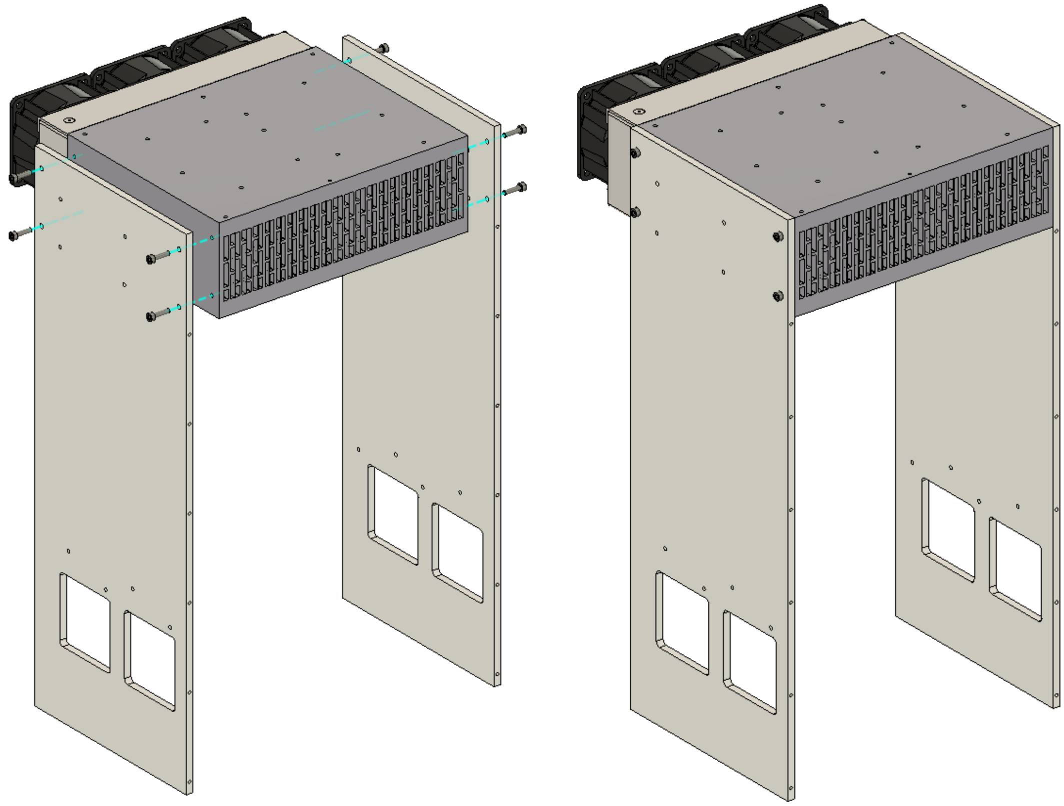
Goal: Mount the power supplies to their holders using 4 x M4x8mm screws. Mount the holders to the backside using 4 x M2.5x12mm screws. Mount all four power supplies. Prepare each output cables such that they can reach their respective LED driver.





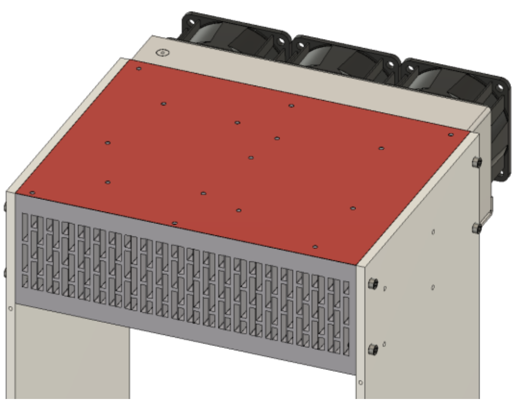
Step 8: Driver Sides

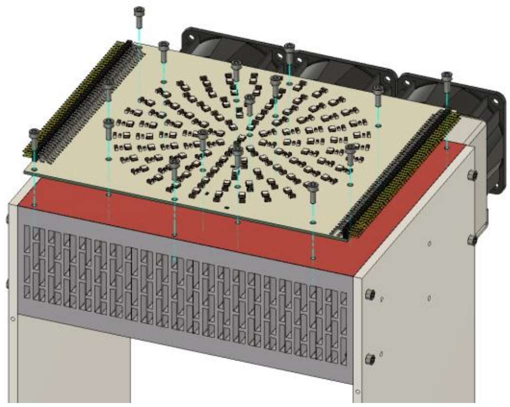
Goal: Mount the driver holders to the heat sink using 8 x M3X12mm screws.

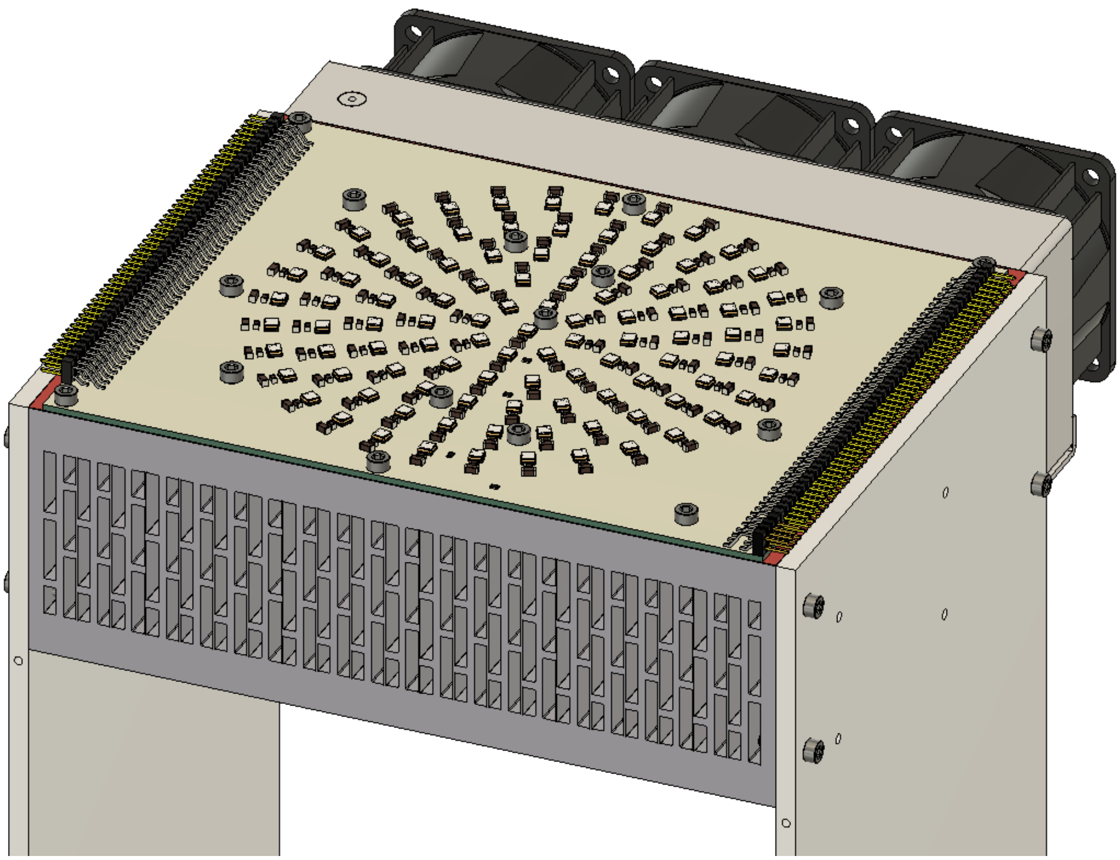


Step 9: Thermal paste and LED array

Goal: Apply a thin continuous layer of thermal paste. Mount the LED array using 15 x M3x6mm screws.

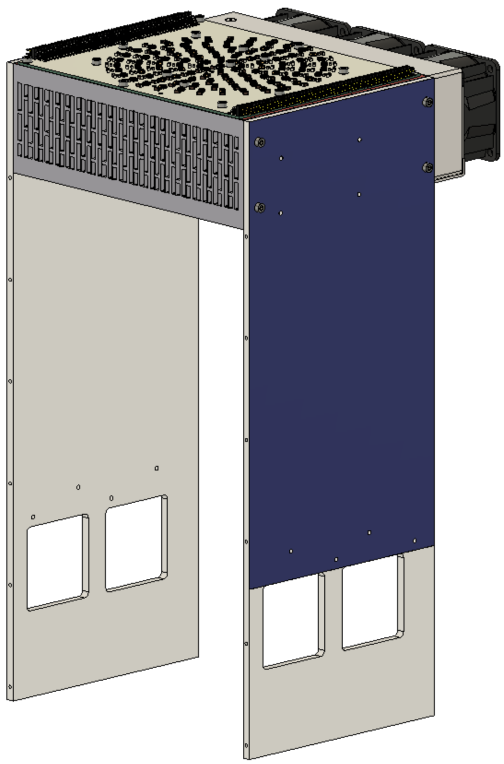
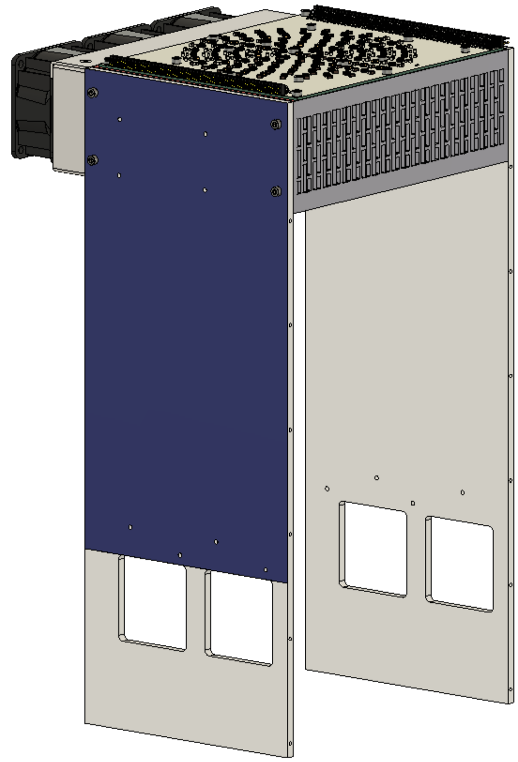
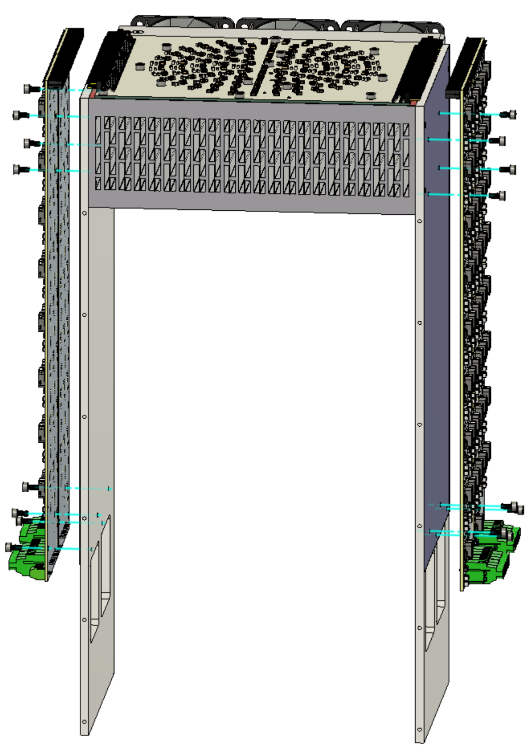
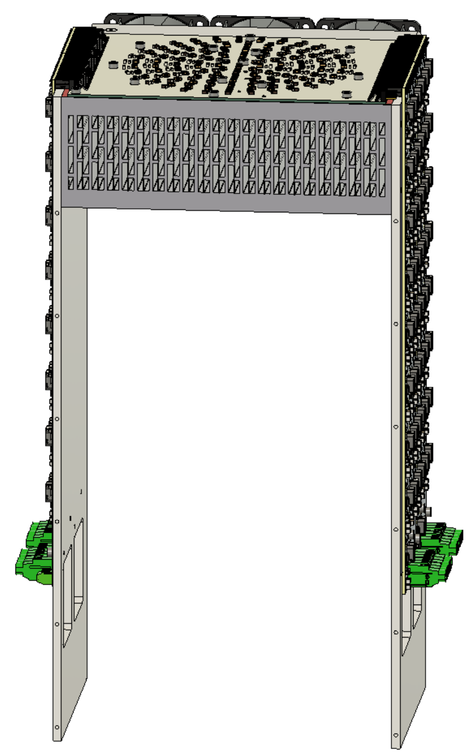






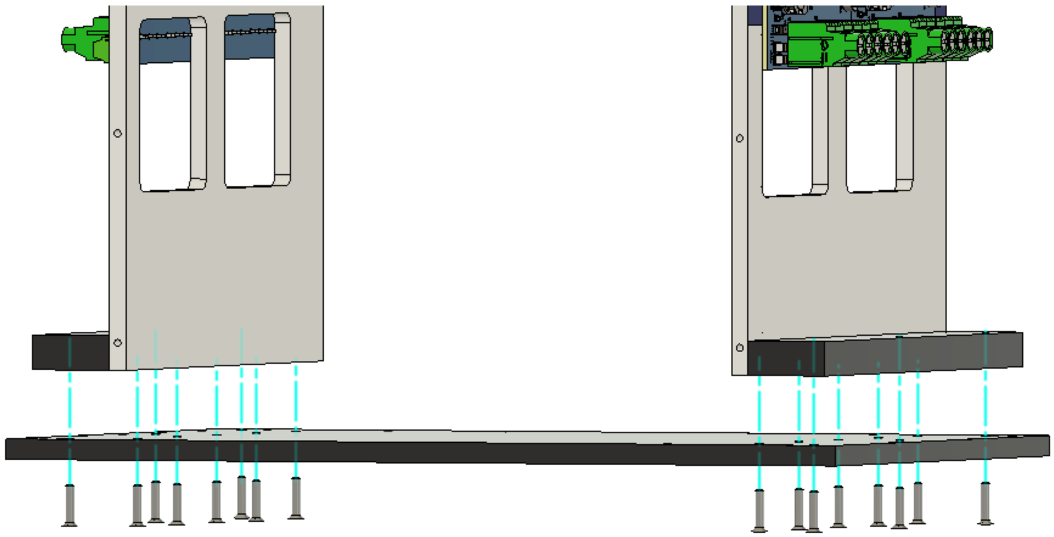
Step 10: Thermal tape and driver mount

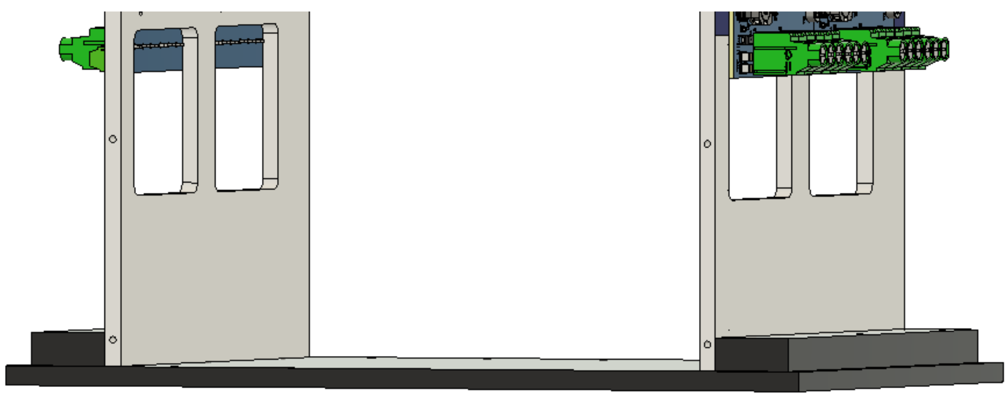
Goal: Apply thermal tape to the driver sides or the LED drivers. Mount the four LED drivers using 4 x M3x6mm screws each.

Step 11: Driver sides and base

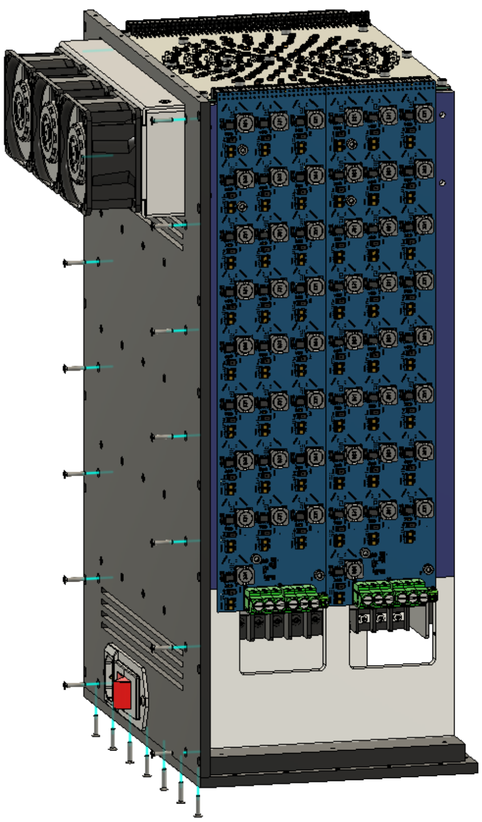
Goal: Mount the driver sides to the base using 10 x M2.5x12mm screws. Attach spacers on both sides using 6 x M2.5x12mm screws.

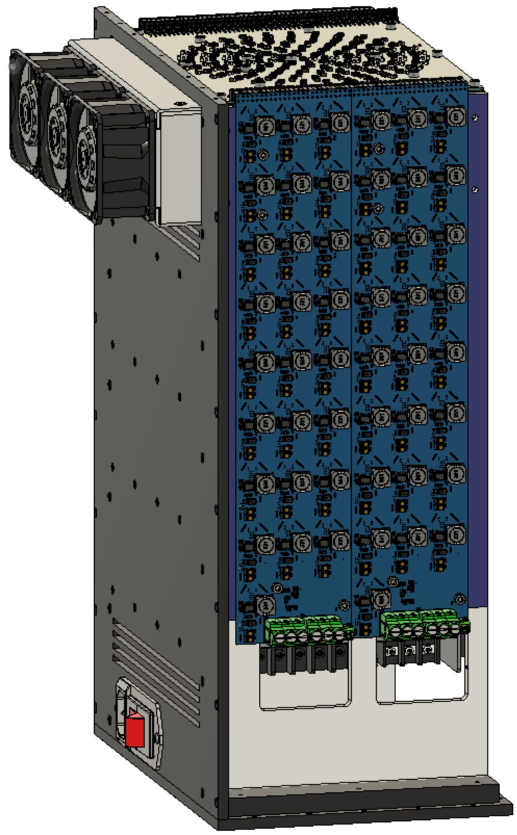




Step 12: Mount the back

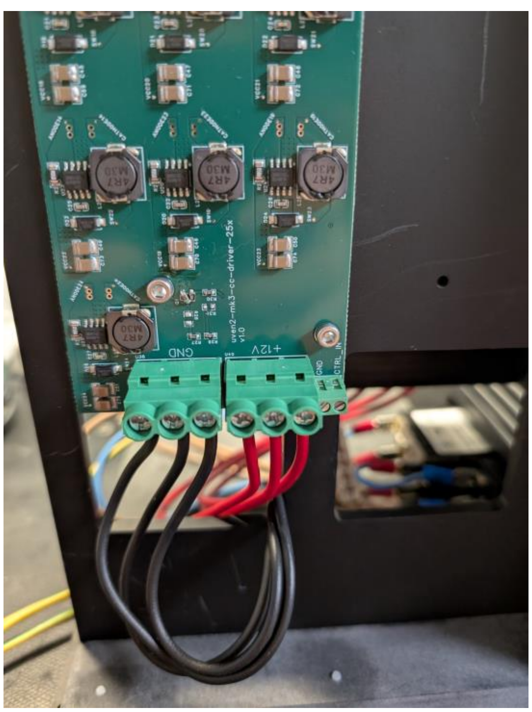
Goal: Mount the back to the driver sides and base using 21 x M2.5x12mm screws.

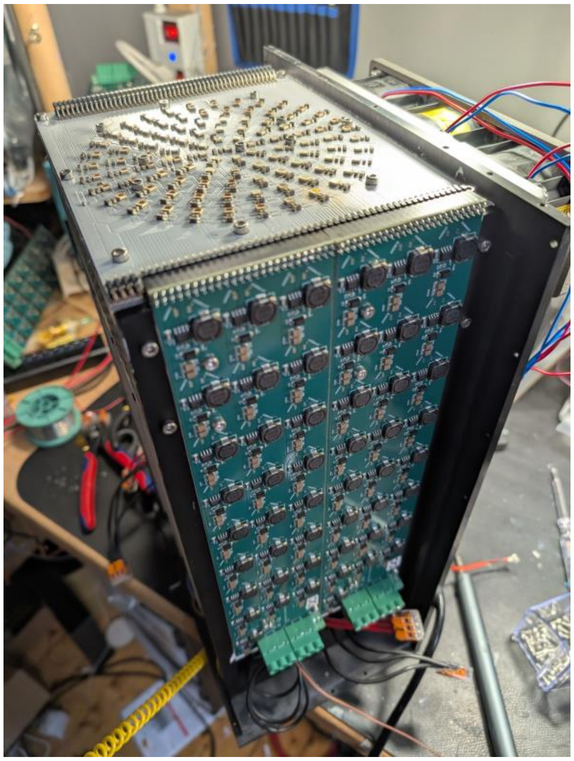


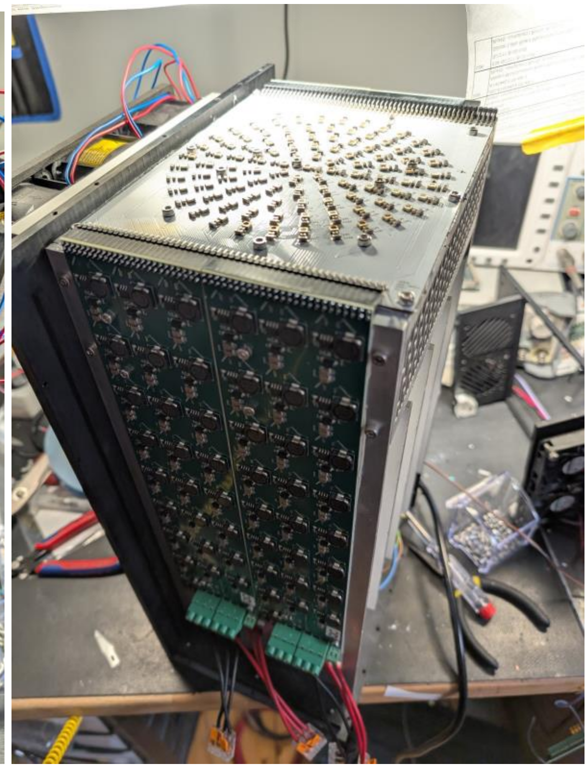


Step 13: Connect 12V supplies

Goal: Connect one 12V output to each LED driver using the screw terminals.

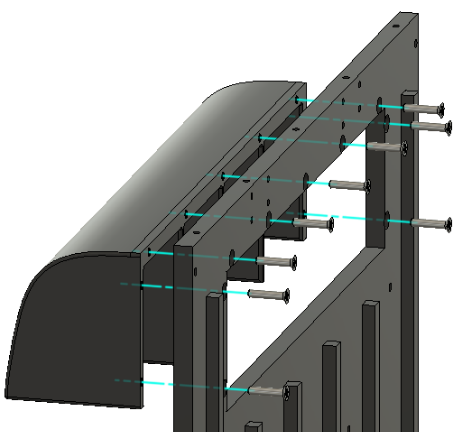


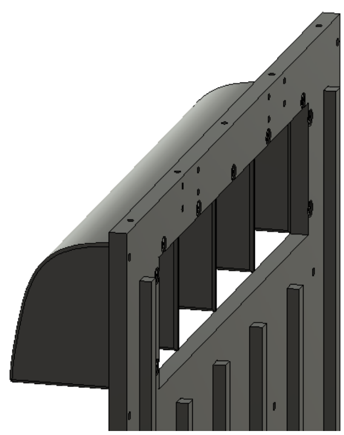




Step 14: Exhaust shield

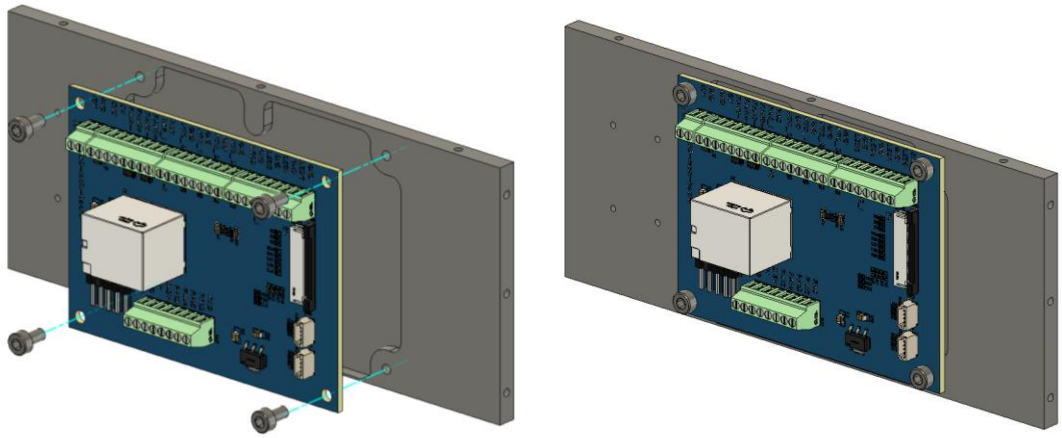
Goal: Mount exhaust shield to front using 9 x M2.5x12mm screws.





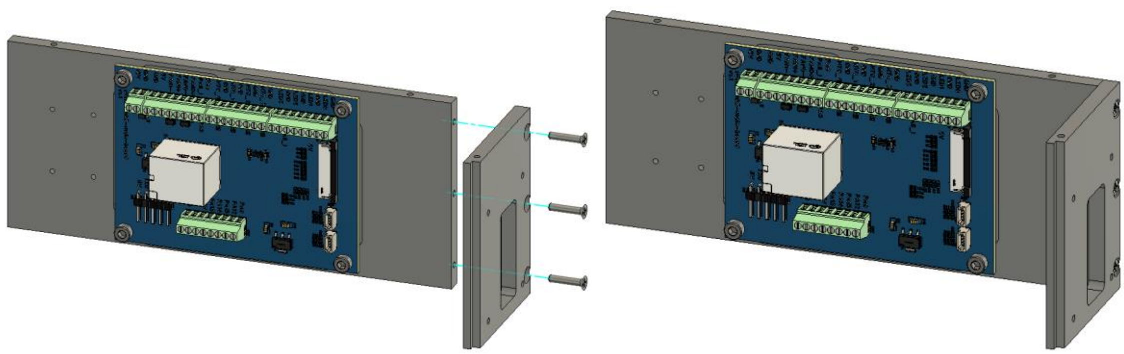
Step 15: Main board

Goal: Mount the main board to the side using 4 x M3x6mm screws.



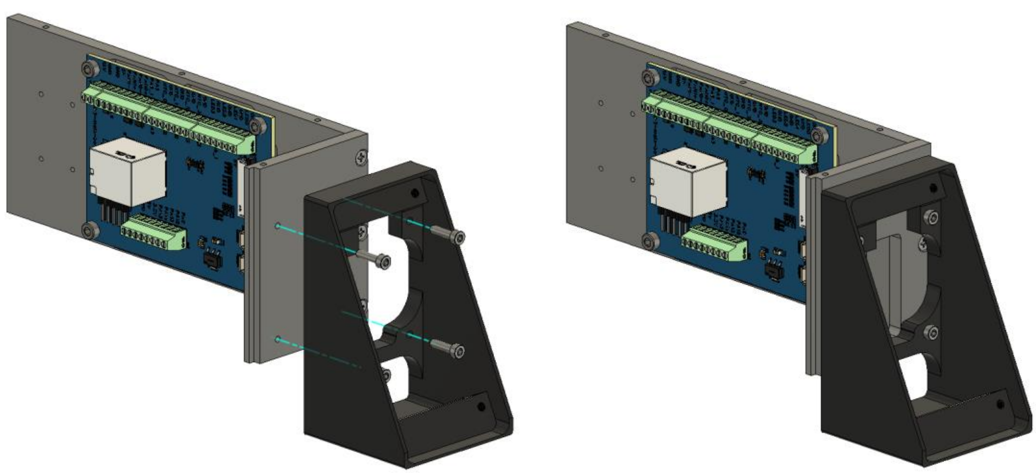
Step 16: Front right

Goal: Mount the front right part using 3 x M2.5x12mm screws.



Step 17: Front panel

Goal: Mount the front panel bottom using 4 x M2.5x10mm screws.

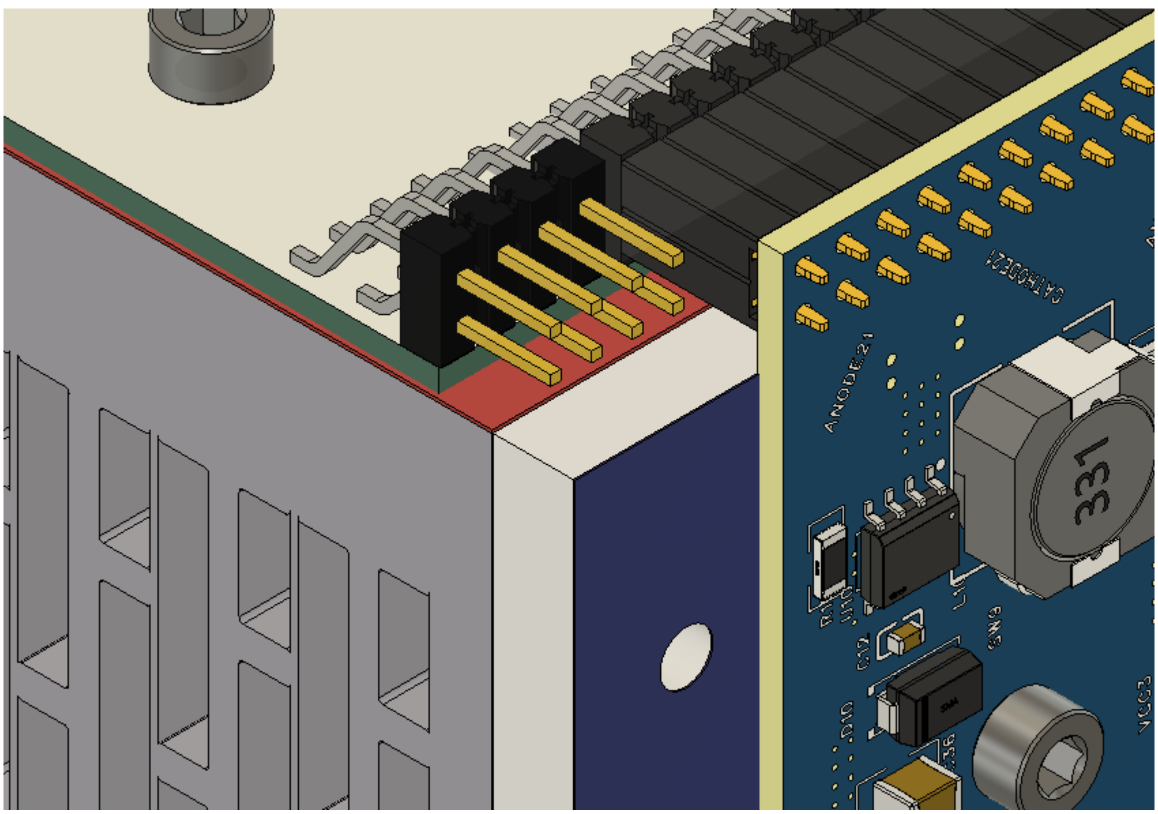


Step 18: Display

Goal: Mount the display using 4 x M3x12mm screws.

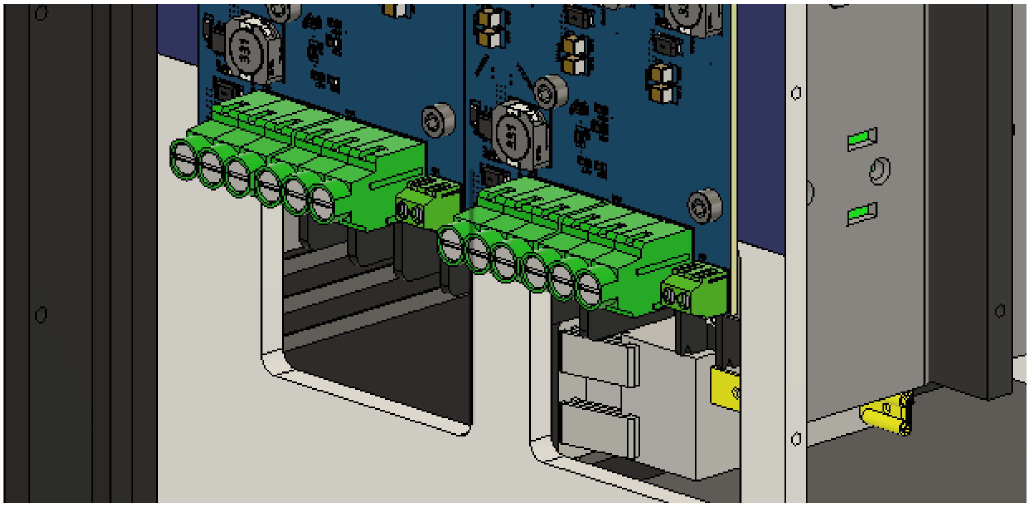
Step 19: Thermistors

Goal: Connect the Thermistors to the main board.



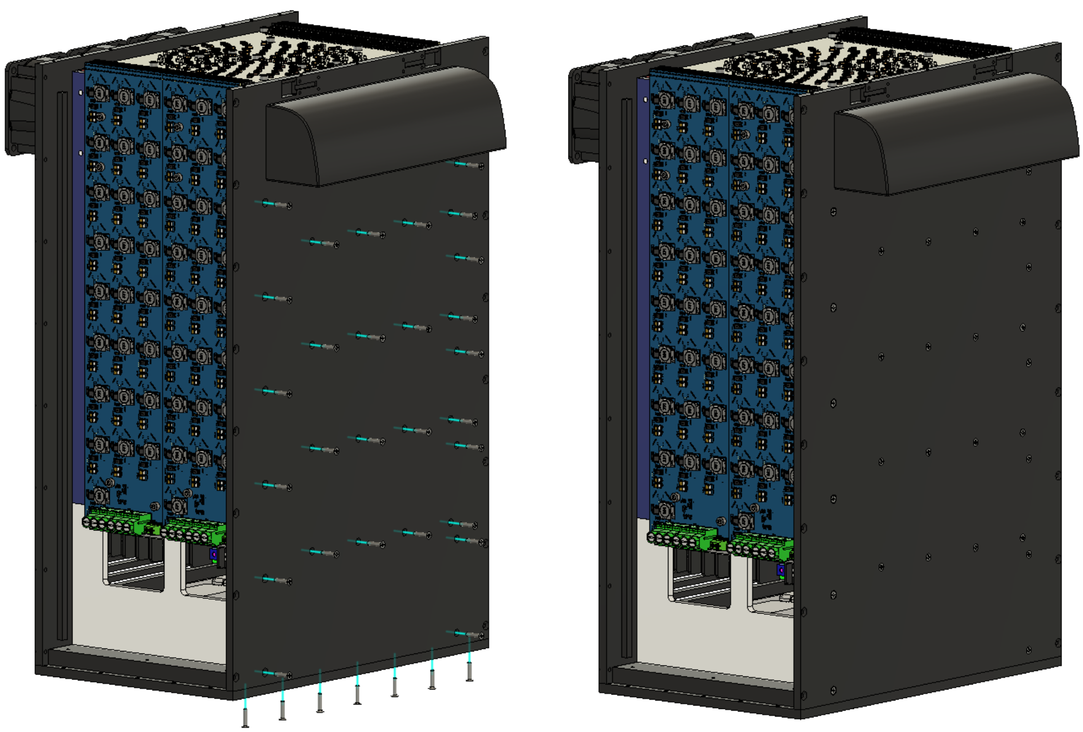
Step 20: LED driver connection

Goal: Connect each LED driver to the main board.



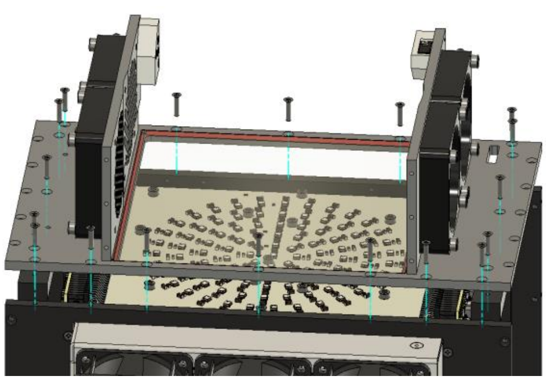
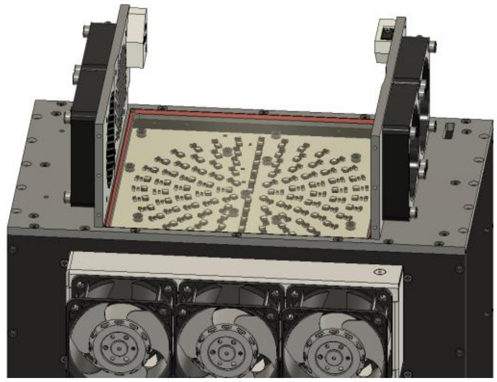
Step 21: Front

Goal: Mount the front panel bottom using 35 x M2.5x12mm screws.



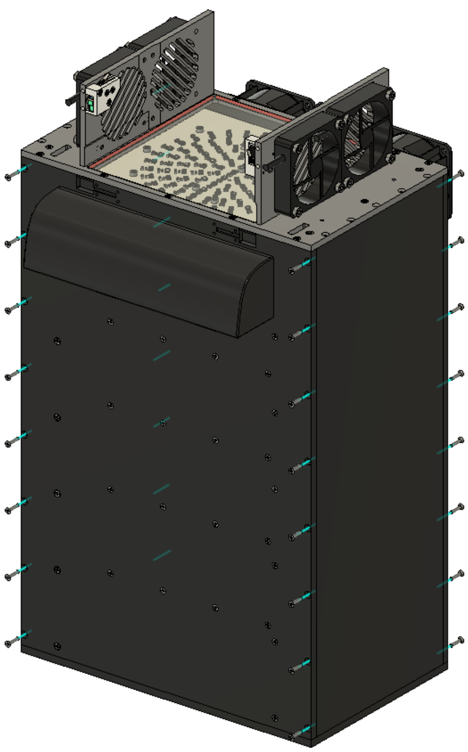
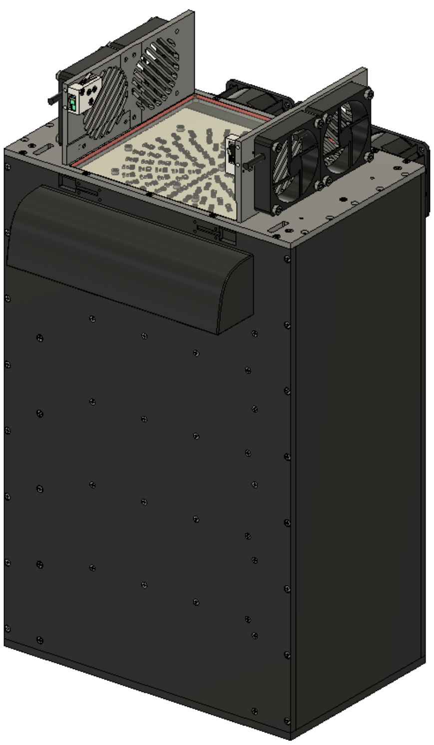
Step 22: Chamber

Goal: Mount the chamber base to the top using 18 x M2.5x12mm screws

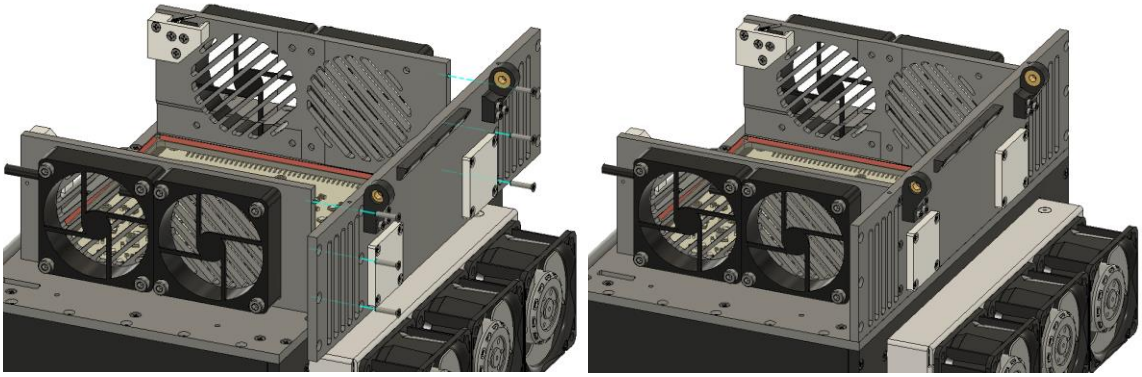
Step 22: Driver sides

Goal: Mount the driver covers using 32 x M2.5x12mm screws. (top and bottom screws not necessary to assemble).

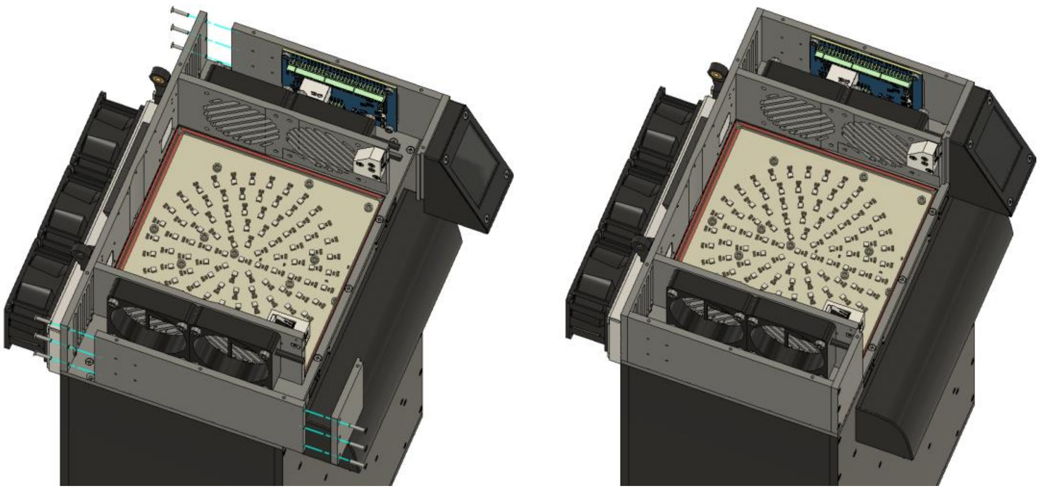
Step 23: Chamber back

Goal: Mount the chamber back using 6 x M2.5x12mm screws.



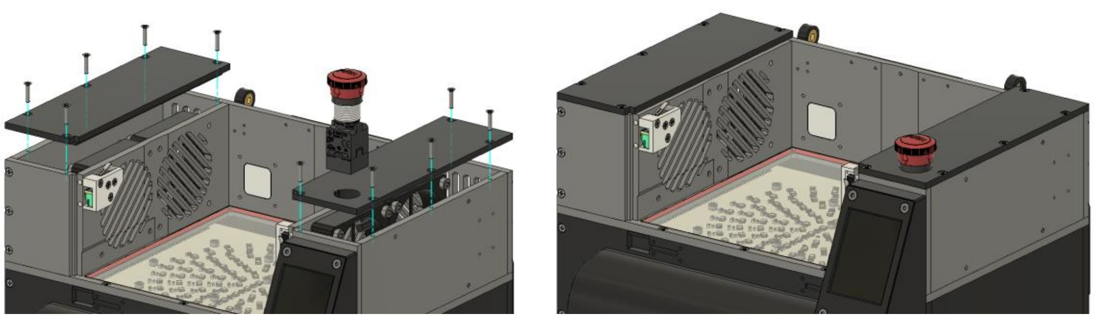
Step 22: Chamber sides

Goal: Mount the chamber sides using 9 x M2.5x12mm screws



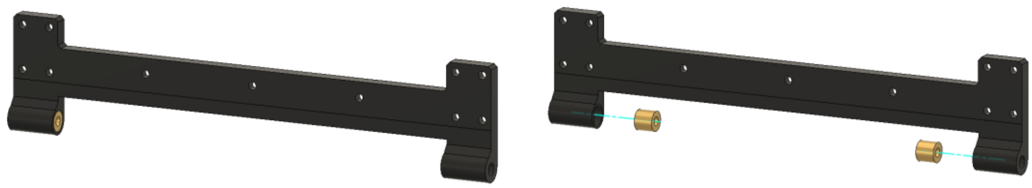
Step 22: Side covers

Goal: Mount the emergency button. Then close the covers using 10 x M2.5x12mm screws.



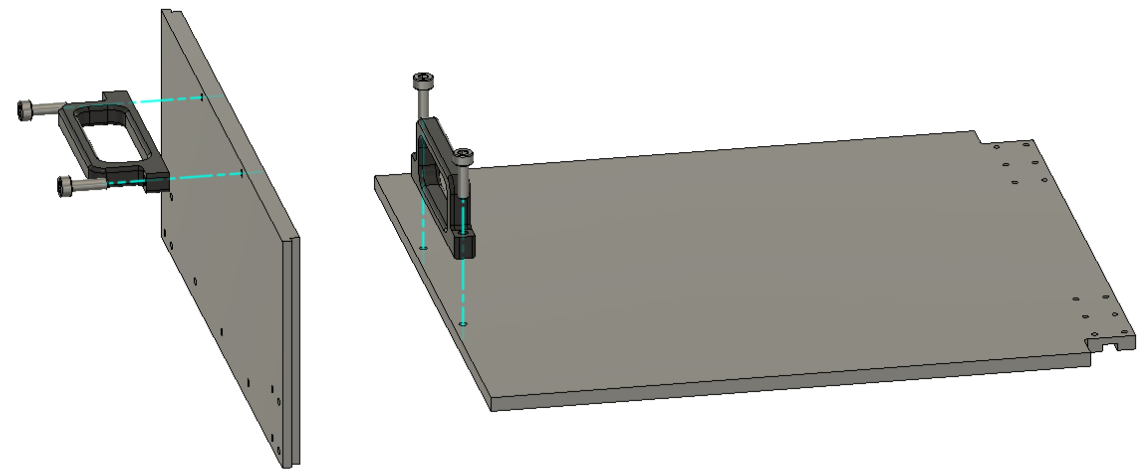
Step 22: Front hinge bushings/bearings

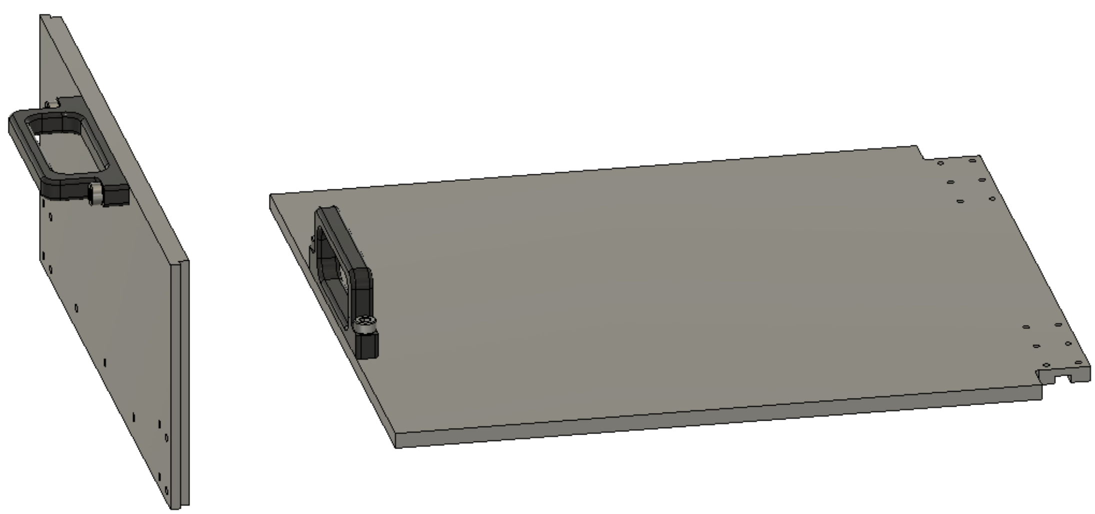
Goal: Press-fit the bushings/bearings in the front hinge.



Step 22: Handles

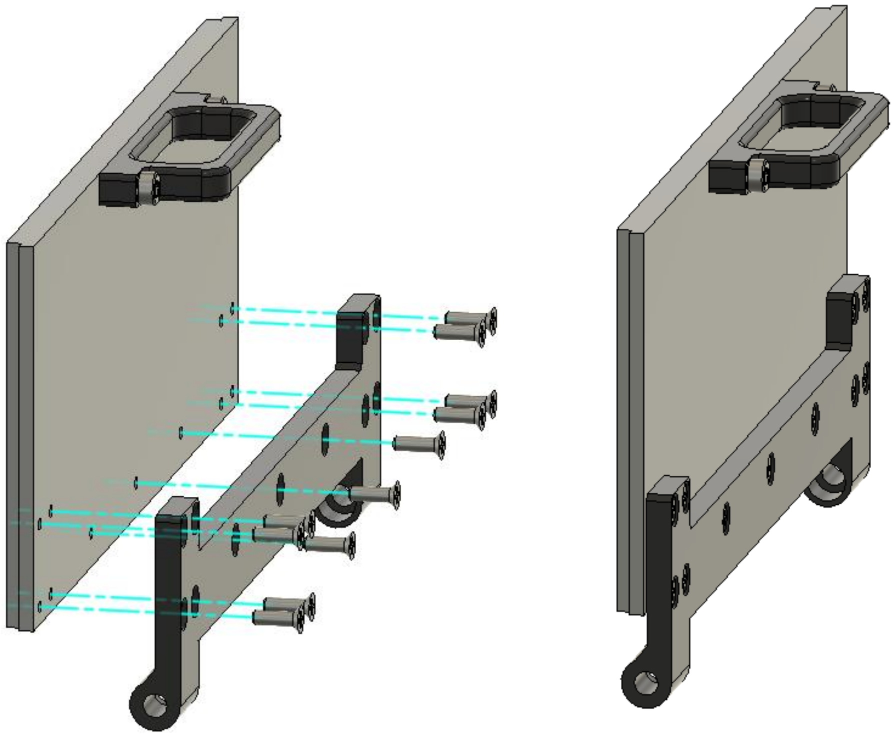
Goal: Mount the handles to the lid and the chamber_front using 4 x M3x8mm screws.





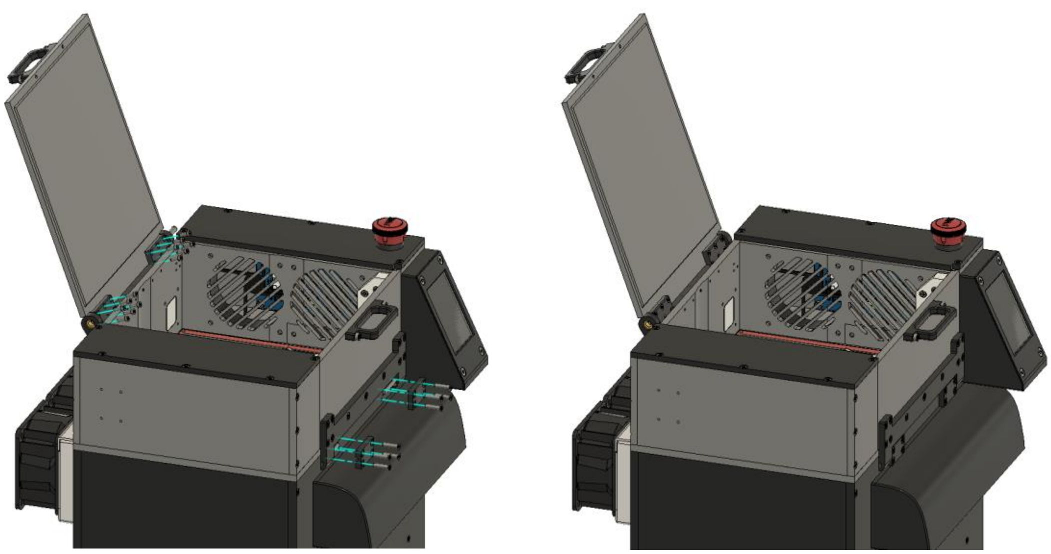
Step 22: Front hinge

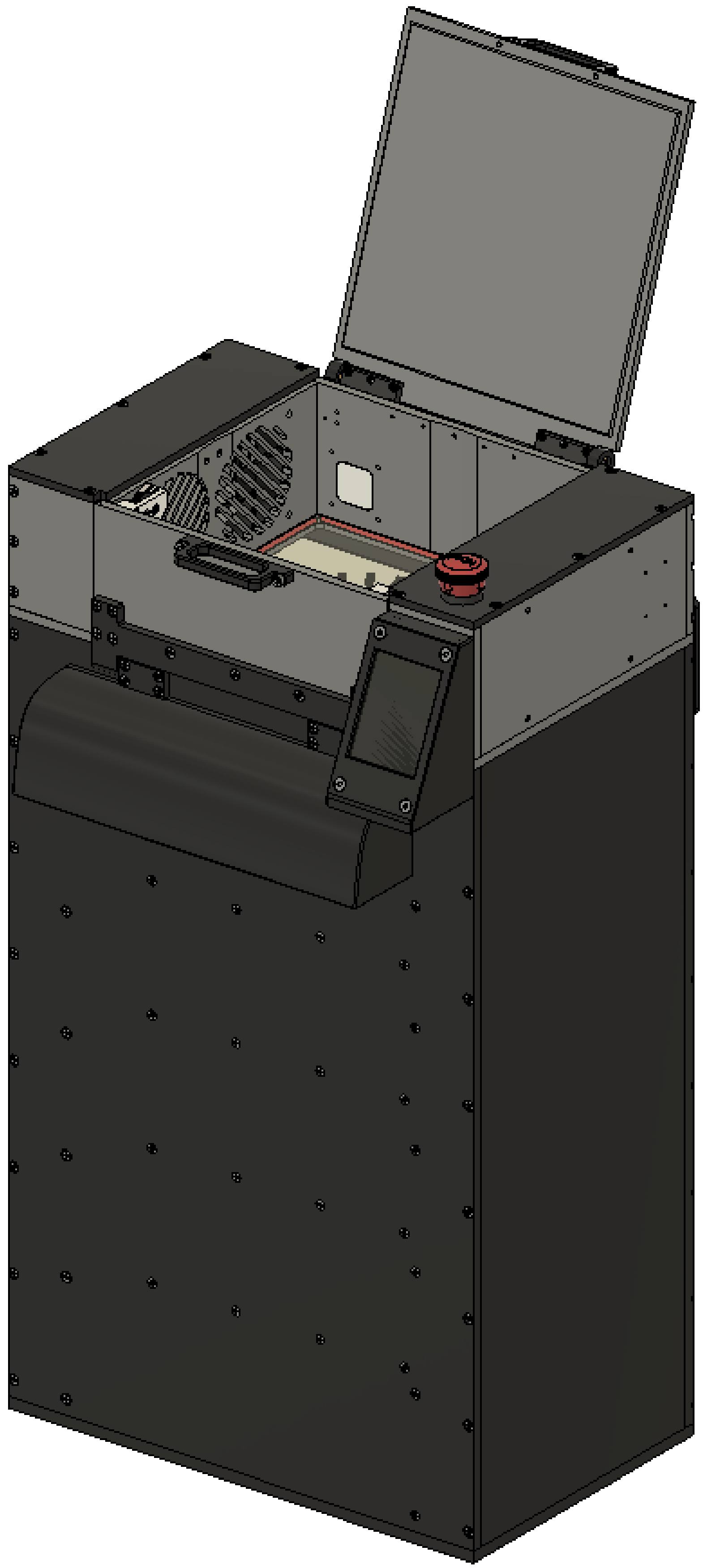
Goal: Mount the hinge to the chamber front using 11 x M2.5x8mm screws.



Step 22: Lid and front

Goal: Mount the lid using 2 x 4x30mm dowel pins and 12 x M2x6mm screws. Mount the chamber front using 2 x 3x30mm dowel pins and 12 x M2.5x12mm screws.





7. Electrical / Wiring Instructions

Wire ID	From	To	Color	Notes

8. Final Assembly

1. [...]
2. [...]
3. [...]

9. Functional Checks

Test ID	Description	Expected Result	Pass/Fail

10. Revision History

Version	Date	Author	Description